

Cellular Respiration, Glycolysis & ATP Synthesis

Academic Lecture Notes & Study Companion

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Core Conceptual Takeaways

- **Universal Energy Currency:** Cellular respiration oxidizes glucose to generate Adenosine Triphosphate (ATP) to drive thermodynamically unfavorable biochemical reactions.
- **Three-Stage Pathway:** Glycolysis (Cytoplasm) → Krebs Cycle (Mitochondrial Matrix) → Oxidative Phosphorylation & ETC (Inner Mitochondrial Membrane).
- **Chemiosmotic Coupling:** High-energy electrons from NADH and FADH₂ establish a transmembrane proton gradient (proton motive force) that drives ATP Synthase rotation.

1 Cellular Respiration and Bioenergetics

Cellular respiration is the foundational catabolic process by which aerobic organisms extract stored potential chemical energy from covalent bonds in glucose molecules.



High-Yield Mnemonic

O-I-L • R-I-G: Oxidation Is Loss of electrons; Reduction Is Gain of electrons.

2 Stage 1: Glycolysis (Cytoplasmic Sugar Splitting)

Glycolysis Characteristics

- **Subcellular Location:** Cell Cytoplasm (Cytosol).
- **Oxygen Dependency:** Anaerobic (requires zero oxygen).
- **Substrate:** 1 molecule of 6-carbon Glucose (C₆H₁₂O₆).
- **End Products:** 2 molecules of 3-carbon Pyruvate, 2 Net ATP (via substrate-level phosphorylation), and 2 NADH.

The pathway operates in two functional phases: the *Energy Investment Phase* (consumes 2 ATP to phosphorylate intermediates) and the *Energy Payoff Phase* (generates 4 ATP and 2 NADH), netting **+2 ATP**.

3 Stage 2: The Krebs Cycle (Citric Acid Cycle)

Pyruvate is translocated into the **mitochondrial matrix** via the pyruvate translocase transporter, where the enzyme *pyruvate dehydrogenase* decarboxylates it into **Acetyl-CoA**, releasing CO₂ and yielding 1 NADH per pyruvate.

3.1 Cycle Dynamics in the Mitochondrial Matrix

Acetyl-CoA (2C) combines with oxaloacetate (4C) to synthesize citrate (6C). Through sequential oxidation-reduction steps:

- 2 carbons are fully oxidized and discharged as waste CO₂.
- Per glucose (2 turns of the cycle): yields **6 NADH**, **2 FADH₂**, and **2 ATP** (or GTP).

4 Stage 3: Oxidative Phosphorylation & The Electron Transport Chain (ETC)

Located along the folded cristae of the **inner mitochondrial membrane**, multi-protein complexes I through IV extract electrons from electron shuttles:

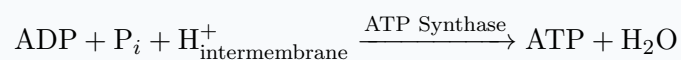
1. **Complex I (NADH Dehydrogenase):** Accepts electrons from NADH, pumping H⁺ protons from matrix to intermembrane space.
2. **Complex II (Succinate Dehydrogenase):** Accepts electrons from FADH₂ (does not pump protons).
3. **Complex III (Cytochrome *bc*₁):** Shuttles electrons through cytochrome *c*, pumping protons.
4. **Complex IV (Cytochrome *c* Oxidase):** Transfers electrons to molecular oxygen (O₂), the **terminal electron acceptor**, reducing it to form metabolic water (H₂O).

5 Chemiosmosis and ATP Synthase

The active pumping of protons creates a steep electrochemical gradient across the inner mitochondrial membrane, termed the **Proton Motive Force (PMF)**.

ATP Synthase Catalytic Turbine

Protons cannot diffuse through the lipid bilayer and instead flow down their gradient back into the matrix exclusively through the *F*₀ rotor channel of **ATP Synthase**. This proton flux induces conformational rotation of the *F*₁ catalytic subunit, phosphorylating ADP and inorganic phosphate (P_{*i*}) into ATP:



This chemiosmotic mechanism yields approximately **26 to 28 ATP**, accounting for over 90% of total cellular ATP.

High-Yield Mnemonic

G-K-E: Stages of Respiration: **G**lycolysis (Cytoplasm) → **K**rebs (Matrix) → **E**TC (Inner Membrane).

Stage	Subcellular Location	ATP Yield	Electron Shuttles
Glycolysis	Cytoplasm (Cytosol)	2 ATP (net)	2 NADH
Pyruvate Oxidation	Mitochondrial Matrix	0 ATP	2 NADH
Krebs Cycle	Mitochondrial Matrix	2 ATP	6 NADH, 2 FADH ₂
Oxidative Phosphorylation	Inner Mitochondrial Membrane	≈ 26–28 ATP	All carriers oxidized
Total Aerobic Net Yield		30–32 ATP	